

**Lecture 20 — 16/10/2025***Lecturer: Prof Minati De**Scribe: Team 20***Scribed by:**

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**Recap of Last Lecture**

- Formal difference between *Turing-recognizable* (semi-decidable) and *decidable* languages.
- Universal acceptance language  $A_{\text{TM}}$  is recognizable but not decidable; this is our seed for many reductions.
- Not every language is recognizable: counting/diagonalization shows there are more languages than TMs.
- A language is decidable iff both it and its complement are recognizable.
- We will use *mapping reductions* to transfer undecidability between problems.

**Overview of this lecture**

- Revisit *recognizability* and *decidability*.
- Canonical language:  $A_{\text{TM}} = \{\langle M, w \rangle \mid w \in L(M)\}$ .
- There exist languages that are *not* Turing-recognizable.
- Use *reductions* to prove undecidability of new problems.

**1 Recognizability vs. Decidability of  $A_{\text{TM}}$** **1.1 Existence of non-recognizable languages**

- The set of all TMs is *countable*, while the set of all languages over  $\Sigma$  is *uncountable*.
- Hence, some languages are not Turing-recognizable.
- In particular, since  $A_{\text{TM}}$  is recognizable but undecidable,  $\overline{A_{\text{TM}}}$  cannot be recognizable (otherwise both  $L$  and  $\overline{L}$  recognizable  $\Rightarrow L$  decidable).

## 2 Reductions: Template to Prove Undecidability

To prove a language  $A$  is not decidable:

1. Start from a *known* undecidable problem (e.g.,  $A_{\text{TM}}$  or  $\text{HALT}_{\text{TM}}$ ).
2. Assume, for contradiction, a decider for the *target* problem.
3. **Using** that decider, *construct* a TM that would decide the known undecidable problem.
4. Contradiction  $\Rightarrow$  target problem is not decidable.

In this lecture, we focus on solving three such problems.

## 3 Problem 1: Halting Problem

**Halting problem :** Input:  $\langle M, w \rangle$ . Question: “Does  $M$  halt on  $w$ ?” Denote

$$\text{HALT}_{\text{TM}} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}.$$

**Claim:**  $\text{HALT}_{\text{TM}}$  is not decidable.

**Proof idea (by contradiction) :** Assume a TM  $H$  decides  $\text{HALT}_{\text{TM}}$ . Build a machine  $R$  that decides  $A_{\text{TM}}$ , contradicting the known result that  $A_{\text{TM}}$  is undecidable.

**Construction of  $R$  :**

On input  $\langle M, w \rangle$ :

1. Run  $H$  on  $\langle M, w \rangle$ .
2. If  $H$  *rejects*, then **reject**.
3. Else, simulate  $M$  on input  $w$ :
  - If  $M$  accepts  $w$ , **accept**.
  - Else ( $M$  rejects  $w$ ), **reject**.

Thus  $R$  decides  $A_{\text{TM}}$  if  $H$  existed, a contradiction. Hence  $\text{HALT}_{\text{TM}}$  is not decidable, and this pattern is our basic reduction gadget.

## 4 Problem 2: Emptiness Testing for TMs

**Empty testing problem :** Given a TM  $M$ , decide whether  $L(M) = \emptyset$ .

$$E_{\text{TM}} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}.$$

**Goal:** Show  $E_{\text{TM}}$  is not decidable by a reduction from  $A_{\text{TM}}$ .

**Proof idea :**

1. Assume a TM  $E$  that decides  $E_{\text{TM}}$ .
2. Using  $E$ , construct a TM  $R$  that decides  $A_{\text{TM}}$  (contradiction).

### Key construction ( $M'$ )

Given  $\langle M, w \rangle$ , define a machine  $M'$  as follows:

$M'$  on input  $x$ :

1. Check if  $x = w$ .
2. If  $x \neq w$ , **reject**.
3. If  $x = w$ , simulate  $M$  on  $w$ :
  - If  $M$  accepts  $w$ , **accept**.
  - If  $M$  rejects  $w$ , **reject**.
  - (If  $M$  loops on  $w$ , so does  $M'$  on  $w$ .)

**Observation :**

$$L(M') = \begin{cases} \{w\}, & \text{if } w \in L(M), \\ \emptyset, & \text{otherwise.} \end{cases}$$

Therefore,  $L(M') = \emptyset$  iff  $\langle M, w \rangle \notin A_{\text{TM}}$ . Now run  $E$  on  $\langle M' \rangle$ :

- If  $E$  *accepts* (i.e.,  $L(M') = \emptyset$ ), then **reject** for  $A_{\text{TM}}$ .
- Else, **accept** for  $A_{\text{TM}}$ .

Hence a decider for  $E_{\text{TM}}$  would yield a decider for  $A_{\text{TM}}$ —contradiction. So  $E_{\text{TM}}$  is not decidable.

## 5 Problem 3: Equivalence Testing for TMs

**Equivalence Testing problem for TMs :** Given two TMs  $M_1, M_2$ , decide whether  $L(M_1) = L(M_2)$ . Define

$$EQ_{\text{TM}} = \{ \langle M_1, M_2 \rangle \mid L(M_1) = L(M_2) \}.$$

**Goal.** Show that  $EQ_{\text{TM}}$  is *not decidable* by a many-one reduction from  $E_{\text{TM}}$ .

*Proof (by contradiction via  $E_{\text{TM}}$ ).* Recall that

$$E_{\text{TM}} = \{ \langle M \rangle \mid L(M) = \emptyset \}$$

is undecidable. Suppose, toward a contradiction, that there exists a decider **EQ** for  $EQ_{\text{TM}}$ .

**Step 1 — Fix a reference machine.** Let  $M^*$  be any TM that *rejects every input*. Then  $L(M^*) = \emptyset$ ; note  $M^*$  is *fixed* and independent of the input  $M$ .

**Step 2 — Define the reduction.** Define the map

$$f : \langle M \rangle \mapsto \langle M, M^* \rangle.$$

For every TM  $M$ , the following equivalences hold:

$$\langle M \rangle \in E_{\text{TM}} \iff L(M) = \emptyset \iff L(M) = L(M^*) \iff \langle M, M^* \rangle \in EQ_{\text{TM}}.$$

**Step 3 — Use EQ to decide  $E_{\text{TM}}$ .** Construct a machine **E** that, on input  $\langle M \rangle$ ,

- (a) computes  $f(\langle M \rangle) = \langle M, M^* \rangle$ ,
- (b) runs **EQ**( $f(\langle M \rangle)$ ),
- (c) if **EQ** accepts (i.e.  $L(M) = L(M^*) = \emptyset$ ) then **accept**, otherwise **reject**.

**Conclusion.** We have built a decider for  $E_{\text{TM}}$ , contradicting its undecidability. Therefore, our assumption that  $EQ_{\text{TM}}$  is decidable must be false. Hence  $EQ_{\text{TM}}$  is undecidable.  $\square$

## 6 Rice's Theorem

**Define** the set  $\mathcal{T} = \{ \langle M \rangle \mid M \text{ is a TM} \}$ . Consider a property  $P \subsetneq \mathcal{T}$  that is:

1. *Non-empty* ( $P \neq \emptyset$ ),
2. *P is proper subset of  $\mathcal{T}$*  (there exists some TM language not in  $P$ ),
3. *Semantic*: for any two TMs  $M_1, M_2$ , if  $L(M_1) = L(M_2)$  then either both languages belong to  $P$  or neither does.

**Then  $P$  is not decidable.**

## References

- [1] P. Linz, *An Introduction to Formal Languages and Automata*, 6th Edition, Jones & Bartlett Learning, 2016.
- [2] M. Sipser, *Introduction to the Theory of Computation*, 3rd ed., Cengage Learning, 2012.